

WHEN DO REUSABLE TEMPORAL PROCEDURES ACTUALLY REPAIR AGENTS?

Anonymous authors

Paper under double-blind review

ABSTRACT

Agents that revisit evolving data can repeat the same procedural mistake even as the underlying values change: using evidence released after a cutoff, confusing publication time with reporting time, or failing to ground a numerical change in contemporaneous documents. A reusable temporal procedure can persist the missing operation, but a standard targeted-versus-generic comparison can be misleading when the generic helper itself harms the original agent. We therefore study an identification problem rather than a skill-learning problem: when does inserting a reusable temporal procedure actually repair a recurring failure? For each frozen endpoint, we compare a targeted procedure (T), a complexity-matched target-blind helper (G), and the original no-skill agent (N); a procedure is a binding repair under this intervention only when T beats both G and N. The extra anchor changes an observed scientific verdict: on one DeepSeek cutoff cell, $T=100\%$ and $G=72\%$, but $N=100\%$, so the apparent +28-point two-arm gain is not a repair. Across 35 frozen endpoints reconstructed from first-party BEA, NOAA, and EIA release histories, positive effects appear selectively in non-ceiling regimes. The strongest held-out cutoff result reaches 100% under T versus 70% under G and 47.5% under N, with endpoint sign test $p = 0.0156$ for T-N; byte-identical procedures also retain direction on compatible future releases, while cross-domain grounding is null and a second model is already at no-skill ceiling on the same EIA tasks. The evidence supports a conditional diagnostic of reusable-procedure bottlenecks. It does not establish universal temporal-skill gains, behavioral neutrality of the executed generic helper, or superiority of a callable skill container over an equivalent retrieval-side implementation.

1 INTRODUCTION

Suppose an agent is asked: *What was the latest value known by the end of April?* A May release is highly relevant to the topic, easy to retrieve, and wrong for the question. On the next period the numbers change, but the required operation does not: filter evidence by the information cutoff. Similar recurring operations arise when an agent must distinguish a publication date from the reporting period it describes, or ground a numerical change in documentary context that was actually available at the time. These are not one-off facts to memorize. They are procedures that should remain useful as institutional releases evolve.

Reusable agent skills are one appealing way to persist such procedures, but the scientific object is the operation rather than the container that stores it. Demonstrating that the operation helps is subtler than comparing a targeted helper with a generic one. If generic assistance distracts the model, then the targeted helper can win while doing no better than the original agent. In that case a two-arm experiment reports a positive “skill effect” even though nothing has been repaired. This is not hypothetical in our data: for one frozen cutoff cell, the targeted procedure scores 100%, the matched generic helper 72%, and the no-skill agent 100%. The targeted-minus-generic contrast is +28 points, but the correct repair verdict is null.

This example separates three questions that are often conflated. *Repair value* asks whether T improves on the original agent N. *Operation specificity* asks whether T beats structured assistance that lacks the target operation: the executed G is target-blind, while a behaviorally neutral G_0 would be a stronger future control. *Container value* asks whether the same operation works better as callable skill state

than as an operation-matched retrieval-side treatment R. The current evidence addresses the first two only under the executed N/G/T design; G_0 and R require new treatments and remain outside the claim.

Our question is therefore: *when does a reusable temporal procedure become a binding repair for a recurring failure?* We call such a case an *intervention-defined causal bottleneck*. For the same frozen endpoint we evaluate three arms: a mechanism-specific targeted procedure (T), a complexity-matched target-blind helper (G), and the original no-skill agent (N). A model-task cell qualifies only when T improves over both G and N under the family outcome frozen before model results are opened. The term *bottleneck* refers only to an operation whose insertion is binding under this executed intervention; it does not claim causal exclusivity, behavioral neutrality of G, or necessity of the callable-skill container.

This definition yields three falsifiable predictions. First, $T > G$ without $T > N$ must be rejected as generic-helper degradation rather than counted as repair. Second, a real procedure bottleneck should be visible only where the original model has headroom; cells already at no-skill ceiling should show no identifiable gain. Third, if the reusable operation rather than endpoint-specific refitting is responsible, byte-identical procedure code should retain its direction on compatible held-out releases, while transfer to an incompatible domain may legitimately be null.

The question is deliberately narrower than recent work on skill learning. Paired trace auditing, no-skill attribution, randomized skill masking, continual skill evaluation, and skill compilation already establish that reusable skills can change agent behavior and that their effects are heterogeneous (Zhou et al., 2026; Dong et al., 2026; Wang et al., 2026b; Guan et al., 2026; Wei et al., 2026; Zheng et al., 2026; Xu et al., 2026). Likewise, RAG systems and temporal retrieval methods can implement time-aware evidence selection upstream (Asai et al., 2024; Yan et al., 2024; Han et al., 2025; Wang et al., 2026a). We therefore do not claim that reusable procedures, temporal filtering, or paired skill causality are new. Our object is the stricter *diagnosis*: whether a predeclared recurring failure is actually repaired by a particular reusable operation rather than merely looking better against a weakened reference.

We study this object using source-native replay built from first-party U.S. Bureau of Economic Analysis (BEA), NOAA Climate Prediction Center, and Energy Information Administration (EIA) release histories (U.S. Bureau of Economic Analysis, 2026; NOAA Climate Prediction Center, 2026; U.S. Energy Information Administration, 2026). The study contains 35 frozen endpoints and 1,326 retained runtime-valid rows. The initial BEA/NOAA replay preserves three mechanism families motivated by TimeSage-EV—cutoff validity, release alignment, and exogenous grounding—while remaining a separate, content-addressed experimental object (Yao et al., 2026). A post-review EIA expansion adds 12 cutoff endpoints using the same targeted and generic implementations byte-for-byte. Repeats test execution robustness; endpoints, not repeats, are the independent scientific units.

The results follow the three predictions rather than a universal “skills help” story. The no-skill anchor first overturns the apparent +28-point DeepSeek cutoff repair described above. In non-ceiling cells, however, targeted procedures beat both controls: DeepSeek documentary grounding reaches 60% versus 20%/20%; held-out EIA cutoff filtering reaches 100% versus 70%/47.5%, with an endpoint sign test of $p = 0.0156$ for T-N; and held-out Kimi alignment reaches 12/12 versus 4/12/7/12. Transfer is conditional rather than universal: compatible future-release tests retain the expected direction, cross-domain grounding is null, and the mini model is already at no-skill ceiling on the same held-out EIA cutoff endpoints.

Contributions. Our contribution is an identification contract for reusable-procedure repair, not a new skill optimizer. We (i) define a three-arm procedure-bottleneck estimand that requires improvement over both generic assistance and the original agent; (ii) show an observed two-arm false positive in which the no-skill anchor changes the scientific verdict; and (iii) test the same frozen procedures across future releases while preserving ceiling and incompatible-domain nulls. We additionally make the identification boundary explicit: the executed contrasts estimate operation value under the frozen harness, whereas a behaviorally neutral helper and an operation-matched retrieval treatment are separate, unexecuted estimands. The resulting claim is intentionally conditional on procedure $\times$ model $\times$ task regime.

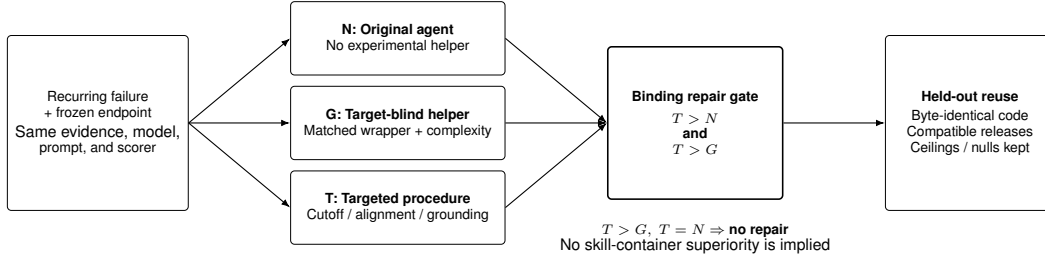


Figure 1: Three-arm procedure-bottleneck test. The same frozen endpoint is evaluated as the original no-skill agent (N), with a complexity-matched target-blind helper (G), and with the targeted reusable procedure (T). A binding repair requires T to beat both N and G; unchanged-code held-out replay separately tests reuse.

2 FROM RECURRING FAILURE TO A PROCEDURE BOTTLENECK

The intuition is to treat a reusable procedure as an intervention, not as a label attached to a successful run. The original agent must first provide the reference behavior: if it already succeeds, a targeted helper cannot be credited with repairing that endpoint. A target-blind helper then tests whether the gain is specific to the nominated operation rather than to receiving some structured assistance. Only after both comparisons are visible do we ask whether the same unchanged operation retains its direction on future compatible endpoints.

Let endpoint i specify a question x_i , a cutoff-valid evidence package E_i , a model configuration M_i , and a family-specific binary success rule $Y_i \in \{0, 1\}$. We intervene only on the reusable procedure exposed to the frozen harness. The three executed conditions are N (no experimental helper), G (a complexity-matched target-blind helper), and T (the mechanism-specific targeted procedure). All other endpoint inputs and scoring rules are held fixed.

For a set of frozen endpoints $\mathcal{I}$ within one model-task regime, we report two empirical contrasts,

$$\hat{\Delta}_{T-N} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} (\bar{Y}_i(T) - \bar{Y}_i(N)), \quad \hat{\Delta}_{T-G} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} (\bar{Y}_i(T) - \bar{Y}_i(G)), \quad (1)$$

where $\bar{Y}_i(a)$ averages repeated executions of the same endpoint under arm a . Repeats characterize execution robustness and do not increase the number of independent endpoint units. The two contrasts have different jobs: $\hat{\Delta}_{T-N}$ asks whether inserting the procedure changes the original agent, whereas $\hat{\Delta}_{T-G}$ asks whether the targeted operation outperforms the executed target-blind helper. Neither contrast establishes that G is behaviorally neutral or that the callable-skill container itself is necessary.

Definition 1 (intervention-defined procedure bottleneck). A targeted procedure is a *binding repair under the executed three-arm intervention* for a frozen model-task regime only when it improves the registered family outcome over both the original agent and matched generic assistance, i.e., when the endpoint evidence supports $\hat{\Delta}_{T-N} > 0$ and $\hat{\Delta}_{T-G} > 0$. A positive targeted-minus-generic contrast with $\hat{\Delta}_{T-N} \leq 0$ is explicitly classified as *no repair*. This definition identifies the value of inserting the operation under the frozen intervention; it does not imply causal exclusivity, behavior-neutrality of the generic helper, or superiority over another implementation of the same operation.

The design targets three failure modes that recur as release content changes.

2.1 CUTOFF VALIDITY

The cutoff procedure receives a query date and candidate evidence timestamps and returns only evidence admissible by that date. It addresses future leakage: a later release may be highly relevant yet invalid for an earlier information set. The matched generic helper can validate evidence-package structure but is forbidden from comparing release dates with the cutoff.

2.2 RELEASE ALIGNMENT

The alignment procedure maps a source release to the reporting period represented by its content and verifies the selected time-series row against that period. It addresses cases in which publication time and reporting time differ. The matched generic helper can inspect document structure but cannot perform the date-to-period mapping.

2.3 EXOGENOUS GROUNDING

The grounding procedure links a numerical change to first-party documentary evidence available before the cutoff. It operates over the full frozen candidate-span region rather than a gold-only pool and returns evidence tied to the requested attribution slots. Its generic helper exposes document structure without the mechanism-specific contextual event selector.

No targeted procedure contains scenario-specific answers, gold labels, or endpoint-specific fitted values. Source hashes are frozen before execution, and unchanged implementations are reused on held-out periods. This gives three direct tests of the bottleneck interpretation: a two-arm false positive must be rejected by N ; gains should disappear at no-skill ceiling; and compatible held-out releases should preserve direction without rewriting the procedure.

3 RELATED WORK

Causal skill attribution. Paired trace auditing, matched/no-skill comparisons, and randomized skill masking already show that exposing an agent to a skill can causally change behavior and that per-skill effects are heterogeneous (Zhou et al., 2026; Dong et al., 2026; Wang et al., 2026b). We therefore do not claim paired skill causality, no-skill attribution, or heterogeneous skill effects as new. Our residual question is stricter: for a *predeclared recurring failure*, does one targeted operation repair the original agent while also outperforming matched generic assistance? The observed $T > G$ but $T = N$ cutoff cell shows why the second comparison changes the scientific verdict rather than merely adding another baseline.

Skill evolution and reusable procedures. ContinualSkillBench, Evo-Harness, SkillProx, HyperSkill, and Anything2Skill study continual skill maintenance, compilation, optimization, structured skill memory, and conversion of external knowledge into reusable procedures (Guan et al., 2026; Wei et al., 2026; Zheng et al., 2026; Xu et al., 2026; Pan et al., 2026). These works make it untenable to claim that procedure reuse, cross-task skill accumulation, or external-knowledge-to-skill compilation is itself novel. We instead treat skill acquisition as upstream of our object: once a reusable procedure exists, our three-arm contract asks whether it is a *binding repair* for the recurring failure that motivated its persistence, and unchanged-code replay tests where that diagnosis transfers or disappears at ceiling.

Temporal retrieval and generic assistance. Selective/corrective RAG and temporal retrieval systems already address irrelevant evidence, time-sensitive indexing, and explicit temporal constraints upstream of generation (Asai et al., 2024; Yan et al., 2024; Zhang et al., 2025; Abdallah et al., 2026; Han et al., 2025; Wang et al., 2026a; Liu et al., 2026). A useful cutoff or alignment operation therefore need not be packaged as a callable skill. The strongest same-information alternative to a container claim is an operation-matched retrieval-side treatment that exposes the model to equivalent temporally filtered evidence without callable skill state. We have not executed that comparison, so we do not claim container superiority. Our executed G instead asks a different question: does the targeted operation beat a complexity-matched target-blind helper under the same callable surface? Because G is not guaranteed behaviorally neutral, a second neutral-helper treatment would strengthen that operation-level comparison as well. The residual scientific object in this paper is consequently procedure-bottleneck identification under the executed three-arm intervention, not a new temporal-retrieval algorithm or evidence that skill packaging itself is necessary.

4 TARGETED PROCEDURE INTERVENTION

Each frozen endpoint is replayed under three conditions while its evidence package, model configuration, prompt harness, evaluator, and period remain fixed. The intervention is intentionally simple: compare the original agent with the same agent receiving either a target-blind helper or the one reusable operation predicted by the registered failure mechanism.

Design requirements. The comparison must satisfy three requirements. First, the targeted arm must improve over the original no-skill agent; otherwise there is no repaired failure. Second, it must also improve over a helper with the same callable surface and matched implementation complexity; otherwise the effect may reflect generic assistance rather than the target operation. Third, the endpoint outcome and transfer split must be frozen before the corresponding model outcomes are opened, so neither the mechanism label nor the success criterion can move to fit the result.

The targeted condition injects the registered mechanism-specific procedure. The generic condition injects a frozen target-blind helper under the same callable wrapper. The no-skill condition adds no experimental helper. The scientific unit is the frozen endpoint, not a repeat. We evaluate the two contrasts from Definition 1 jointly: targeted minus no skill asks whether the operation repairs the original agent, while targeted minus generic asks whether that repair exceeds generic helper/code assistance. Neither contrast is sufficient alone.

4.1 MECHANISM CONTRACTS

The cutoff procedure filters candidate evidence by the target cutoff. Its generic control validates package structure without using timestamps. The release-alignment procedure normalizes the reporting period represented by a source release. Its generic control inspects document structure without performing date-to-period mapping. The grounding procedure enriches candidate documentary spans using reusable attribution and outlook cues. Its control can expose document structure but has no contextual event selector.

No targeted procedure contains scenario-specific values, dates, labels, or expected answers. Procedure source hashes are frozen before execution. The targeted and generic source implementations are matched exactly in lexical-token count and AST-node count for each family. This generic control is deliberately *target-blind*, not assumed behaviorally neutral: it controls the callable wrapper and implementation-complexity surface while withholding the temporal operation under test. The no-skill arm is therefore essential for detecting harm introduced by the particular generic helper.

4.2 RESOURCE PARITY AND IDENTIFICATION LAYERS

The experiment measures controlled procedure exposure, not autonomous skill selection. Under the frozen harness, a targeted or generic helper is executed once before the model answer whenever that arm is assigned; the no-skill arm executes no experimental helper. Targeted and generic arms therefore have equal external helper-call budgets and matched source complexity, while T-N deliberately compares against the original unaugmented agent. Our efficiency lane is a parity audit, not a speedup claim: we do not claim lower latency, provider cost, or call count than N.

The identification ladder has four distinct contrasts. T-N asks whether the operation repairs the original agent; T-G asks whether it beats the executed target-blind helper. Two stronger comparisons remain unexecuted: T-G₀ would test the operation against separately validated behavior-neutral extra computation, and T-R would test callable skill state against an operation-matched retrieval-side implementation. Appendix A freezes these prospective treatments and gives the full ladder. Post-hoc invocation, order, length, and date diagnostics close narrower implementation explanations but do not substitute for G₀ or R.

4.3 FROZEN FAMILY OUTCOMES AND SCORER TIMELINE

Before any source-native model outcome is opened, the failure-family contracts define what constitutes success. A cutoff success requires a nonempty set of cited evidence and a selected latest reference that are all available by the target cutoff. A release-alignment success requires the correct

reporting-period identity together with the aligned headline value. A grounding success requires the requested evidence slots to be filled by frozen first-party sentence IDs in the correct slot order.

An auxiliary full-task conjunctive scorer is preserved only as sensitivity analysis because it adds orthogonal output fields and exact surface-form requirements beyond the registered mechanism outcome. The family definitions frozen before execution remain primary; Appendix A records the scorer timeline and representation audit.

The primary decision rule follows Definition 1: $T > G$ with $T \leq N$ is no repair, while $T > N$ without $T > G$ cannot isolate the targeted operation from generic assistance. The executed design therefore supports an operation-level claim only; container value and behavior-neutral-compute value remain separate, unexecuted estimands.

4.4 HELD-OUT TRANSFER AND EVIDENCE LAYERS

The initial replay freezes twelve pre-April endpoints as C3-R and eleven April–May endpoints as held-out C4-R; same-domain and compatible cross-domain labels are fixed before outcomes, and procedure source is unchanged across the split. A separate EIA validation freezes four earlier and eight held-out energy endpoints while reusing T1/G1 byte-identically. DeepSeek is the registered primary track, Kimi a provenance-clean replication selected by a pre-outcome support rule, and the mini-model layer an exploratory capacity stress. Full execution chronology, the failed local-secondary gate, and row-level provenance bindings are reported in Appendix A.

5 EXPERIMENTAL SETUP: SOURCE-NATIVE REPLAY

TimeSage-EV motivates the three recurring mechanism families—cutoff validity, release alignment, and exogenous grounding—and exposes persistent skills to later periods (Yao et al., 2026). Our experiment is not an exact rerun of its unavailable evaluated snapshot. Instead, we reconstruct content-addressed endpoints directly from first-party institutional release histories and intervene on the same reusable operations.

5.1 FROZEN ENDPOINTS AND SCORERS

The initial replay contains 23 endpoints from BEA GDP releases and NOAA-CPC ENSO Diagnostic Discussions (U.S. Bureau of Economic Analysis, 2026; NOAA Climate Prediction Center, 2026): twelve pre-April endpoints form C3-R and eleven April–May endpoints form held-out C4-R. Raw pages are SHA-256 bound, evidence identifiers are opaque, and grounding exposes the full content-region sentence pool rather than a gold-only candidate set. Family scorers are deterministic and frozen before model outcomes; a pre-execution self-test gives 23/23 success on gold outputs and 0/23 on constructed negatives.

A post-review validation adds 12 EIA Weekly Petroleum Status Report cutoff endpoints from a third institutional system (U.S. Energy Information Administration, 2026). EIA is chosen by a structural compatibility rule—recurring first-party releases, explicit publication dates, represented data-ending weeks, and compatibility with the already frozen cutoff interface—before any EIA model outcome is opened. Four endpoints form an earlier diagnostic split and eight April–May endpoints form held-out validation. T1 and G1 are reused byte-for-byte; source manifest, endpoints, scorer, harness, and condition order are frozen before execution. This is informed post-review validation, not an independently preregistered confirmation.

5.2 QUESTIONS AND INFERENCE CONTRACT

RQ1 asks whether adding the no-skill anchor changes a verdict that would look positive under targeted-versus-generic comparison alone. RQ2 asks whether targeted procedures beat both controls specifically in non-ceiling model–task cells. RQ3 asks whether byte-identical procedures retain direction on compatible held-out releases while preserving ceiling and incompatible-domain nulls.

The endpoint is the independent scientific unit. Repeats characterize execution robustness and are averaged within endpoint; they are never counted as additional tasks. For non-ceiling cells we report endpoint win/tie/loss patterns, percentile bootstrap intervals over endpoints, and exact two-sided sign

tests over non-ties when resolution permits. Two- and three-endpoint cells are explicitly descriptive. We do not pool heterogeneous procedures, models, or domains into an omnibus procedure-effect statistic.

5.3 EVIDENCE LAYERS AND PARITY

DeepSeek-V4-Pro is the registered primary track with five repeats. A pre-outcome support rule selects Kimi-K3 as a separately labeled replication; a later hash audit removes one overwritten repeat block without replacement, leaving four provenance-clean repeats. Doubao-Seed-2.0-Mini is an exploratory capacity stress. The EIA layer evaluates DeepSeek and the mini model. In total, the retained source-native evidence contains 1,326 runtime-valid rows over 35 distinct endpoints from BEA, NOAA, and EIA.

Every targeted procedure implementation has a family-matched generic helper with the same callable signature, identical Python lexical-token count and AST-node count, and equal external tool-call budget. Static audits reject generic helpers that access the target temporal operation. Full execution chronology, provenance bindings, and the failed local-secondary runtime gate are reported in Appendix A.

6 RESULTS

We organize the results around the three predictions from Section 2: whether the no-skill anchor changes a two-arm verdict, where a targeted procedure is binding rather than ceiling-limited, and whether unchanged procedures retain direction on compatible future releases. Endpoints are the independent units; repeats measure execution robustness only.

6.1 THE THIRD ARM CHANGES THE SCIENTIFIC VERDICT

The clearest justification for the design is a negative result. On DeepSeek C3-R cutoff, the targeted procedure succeeds on 100% of endpoint-repeat observations and the matched generic helper on 72%. A conventional T-versus-G comparison would therefore report a +28-point benefit. The no-skill agent, however, is also at 100%. Thus $\hat{\Delta}_{T-G} = +28$ points but $\hat{\Delta}_{T-N} = 0$, and Definition 1 classifies the cell as *no repair*.

The frozen G1 helper explains why this is a plausible failure mode for a generic control: it validates schema-compatible evidence but is deliberately forbidden from comparing release dates with the cutoff, so it can foreground both an admissible release and a post-cutoff distractor. T1 performs the missing cutoff operation, but in this cell it only restores the behavior of the original agent. The same pattern appears more weakly in DeepSeek C4-R, while Kimi cutoff is at ceiling. The three-arm design therefore does more than reduce variance: it prevents a generic-arm deficit from being promoted into a procedure-repair claim.

A zero-call r9 audit further shows that T-G is not explained by unequal helper availability: targeted and generic helpers are each forced exactly once in all 442 retained rows of their respective arms, versus 0/442 in no-skill; condition position has no detectable success association in any arm ($p > .65$), and targeted prompts are 18.85 input tokens shorter on average than their exact generic counterparts. These diagnostics close simple invocation/order/length explanations for T-G. They do not turn N into a resource-matched arm, because N executes no helper, and they do not make G behaviorally neutral. The present T-N contrast should therefore be read as incremental value over the original agent, while a neutral-helper candidate G_0 remains the prospective test of whether the targeted operation adds value beyond behaviorally neutral extra computation.

6.2 GROUNDING IS A BINDING REPAIR WHEN THE ORIGINAL AGENT HAS HEADROOM

DeepSeek gives the cleanest primary-track positive case. Across five C3-R grounding endpoints and five repeats, N0 and G3 each succeed on 5/25 endpoint-repeat observations while T3 succeeds on 15/25: 20%, 20%, and 60%, respectively. Hence both contrasts are +40 points, with the same direction in every repeat. At the endpoint level, targeted-minus-generic has two wins, three ties, and

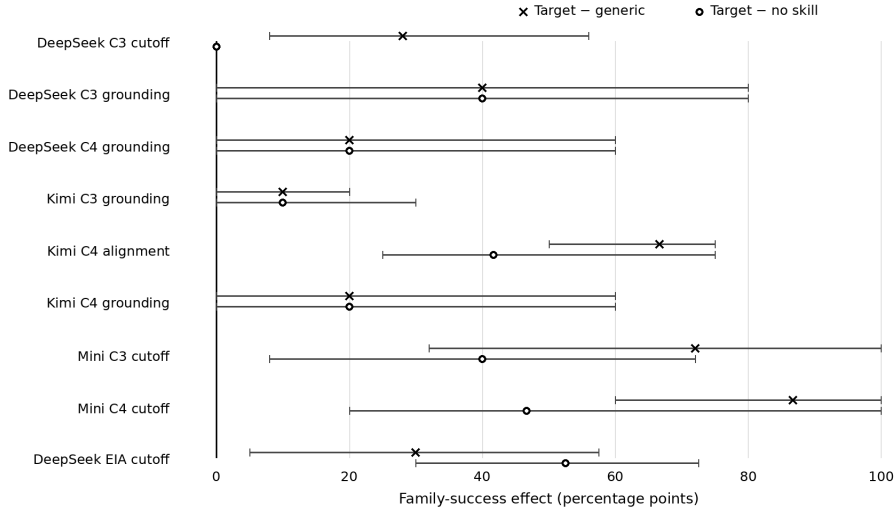


Figure 2: Temporal-procedure effects are regime-dependent. Circles compare targeted with the original no-skill agent; crosses compare targeted with the matched generic helper. Horizontal bars show endpoint-bootstrap 95% intervals. DeepSeek C3 cutoff illustrates the identification point: $T > G$ but $T = N$, so the apparent two-arm gain is not a repair. EIA denotes the frozen post-review energy validation.

Model / split	Family	N0	Generic	Target	Δ_{T-G}	Δ_{T-N}
DeepSeek C3-R Cutoff	Cutoff	100	72	100	+28	0
DeepSeek C3-R Align	Align	100	100	100	0	0
DeepSeek C3-R Ground	Ground	20	20	60	+40	+40
DeepSeek C4-R Cutoff	Cutoff	100	87	100	+13	0
DeepSeek C4-R Align	Align	100	100	100	0	0
DeepSeek C4-R Ground	Ground	40	40	60	+20	+20
Kimi repl. C3-R Cutoff	Cutoff	100	100	100	0	0
Kimi repl. C3-R Align	Align	100	100	100	0	0
Kimi repl. C3-R Ground	Ground	30	30	40	+10	+10
Kimi repl. C4-R Cutoff	Cutoff	100	100	100	0	0
Kimi repl. C4-R Align	Align	58	33	100	+67	+42
Kimi repl. C4-R Ground	Ground	60	60	80	+20	+20

Table 1: Registered DeepSeek and provenance-clean Kimi replication family success (percent). DeepSeek has five repeats; Kimi retains four. Bold entries improve over both controls. Repeats are not treated as independent endpoint samples.

no losses; its endpoint-bootstrap 95% interval is [0,80] points. Kimi shows a smaller C3-R directional gain of 10 points over both controls.

Held-out grounding is directionally consistent but low-resolution. Across all C4-R grounding endpoints, DeepSeek moves from 40% under either control to 60% with T3, and Kimi from 60% to 80%. On the two pre-registered same-domain held-out endpoints, both models show a descriptive 50-point targeted gain over both controls. Cross-domain grounding, however, shows no incremental targeted effect. We therefore interpret the evidence as future-period reuse within the compatible grounding regime, not broad cross-domain portability.

6.3 CUTOFF REPAIR APPEARS IN NON-CEILING REGIMES AND DISAPPEARS AT CEILING

The initial exploratory mini-model stress exposes cutoff headroom that is absent from the primary DeepSeek C3-R cell. Over five repeats, mini-model C3-R cutoff rises from 32% (N0) and 0% (G1) to 72% (T1); held-out C4-R rises from 53% and 13% to 100%. The held-out cross-domain cutoff

subset is 0/5, 0/5, and 5/5 for N0, G1, and T1. Because this layer is exploratory, we use it as a regime probe rather than the primary confirmation.

The complete exploratory matrix is reported in Appendix A; the main text retains only the cutoff cells that bear on the ceiling prediction.

The post-review EIA validation provides a stronger held-out cutoff test without changing the procedure. We freeze 12 new energy endpoints before opening their model outcomes and reuse the original T1/G1 source byte-for-byte. On the eight held-out April–May endpoints, DeepSeek moves from 19/40 under N0 (47.5%) and 28/40 under G1 (70%) to 40/40 under T1 (100%). Targeted-minus-no-skill is positive on seven of eight endpoint means, with one tie and no losses; the endpoint-bootstrap 95% interval is [30,72.5] points and the exact two-sided sign test over the seven non-ties gives $p = 0.0156$. Targeted-minus-generic has four wins and four ties, interval [5,57.5], and sign-test $p = 0.125$.

Crucially, the mini model is already 40/40 under no skill on these same held-out EIA endpoints, leaving no room for T1 to improve. Together with the initial mini-model result, this rejects a monotonic “smaller model, larger skill effect” story. The same cutoff procedure is binding in some model–task cells and unidentifiable at ceiling in others.

6.4 TRANSFER IS SELECTIVE AND EXPLICITLY BOUNDED

Held-out Kimi alignment moves away from the C3-R ceiling: T2 succeeds on 12/12 observations versus 7/12 for N0 and 4/12 for G2. All three independent endpoints favor T over G, but the exact sign test is only $p = 0.25$; DeepSeek remains at ceiling. Elsewhere, T1/G1 are reused byte-identically on EIA and grounding retains direction on compatible future periods, whereas cross-domain grounding is null and the mini model is at no-skill ceiling on EIA. Thus unchanged procedures transfer only in some compatible, non-ceiling regimes. We do not infer a universal temporal-skill effect, a monotonic model-capacity law, or general cross-domain transfer.

7 LIMITATIONS

The unavailable TimeSage-EV evaluated snapshot prevents an exact rerun; our BEA/NOAA/EIA replay is independent mechanistic evidence. Its 35 endpoints are uneven: EIA is cutoff-focused, same-domain grounding transfer has two held-out endpoints, Kimi alignment has three, and cross-domain grounding is null. DeepSeek is primary, Kimi is a provenance-clean replication, the mini model is exploratory, and EIA is post-review validation frozen before outcomes. Repeats measure robustness rather than task sample size. These limits rule out broad cross-domain or model-invariant claims.

Two stronger identification layers remain open. N is the original agent rather than a resource-matched neutral-helper arm, and G is target-blind but not guaranteed behaviorally neutral; a prospectively validated G_0 is required to test value beyond neutral extra computation. No operation-matched retrieval-side treatment was executed, so callable-container value is also unresolved. Appendix A freezes both future treatments; invocation/order/length/date audits do not replace their missing outcomes.

8 CONCLUSION

Reusable temporal procedures can be binding repairs in specific non-ceiling model–task regimes. Requiring T to beat both the original agent and matched target-blind assistance turns one apparent +28-point cutoff effect into no repair, while other grounding, cutoff, and alignment cells remain positive. The lesson is narrower than “skills help”: persistent operations should be promoted only against explicit controls, with neutral-compute and retrieval-container value treated as separate estimands.

REPRODUCIBILITY STATEMENT

The source-native study is content-addressed end to end. The supplementary replication bundle contains the frozen endpoint definitions, prompt harnesses, family scorers, all six executed tar-

geted/generic helper implementations, condition schedules, first-party archived source pages, retained raw result JSON, adjudication code, and a SHA-256 manifest. Appendix A records the executed helper code, the endpoint-level inferential protocol, and the zero-call invocation/order/date-serialization audits. The source-native bundle is independent of the unavailable TimeSage-EV evaluated snapshot; the exact TimeSage replication remains a separately identified external replication target.

AI USE STATEMENT

Generative AI tools were used throughout the research workflow to assist literature triage, research-idea and hypothesis refinement, experimental-design critique, implementation and debugging, data reformatting, result interpretation, internal mock review, and manuscript drafting and editing. AI-assisted experimental artifacts were accepted only after executable checks against frozen task contracts, immutable raw outputs, first-party source material, or deterministic recomputation. AI-generated review feedback had advisory status and did not itself create scientific evidence. The authors reviewed the AI-assisted work and take responsibility for the final paper, claims, code, and supplementary artifacts.

REFERENCES

- Abdelrahman Abdallah, Bhawna Piryani, Jonas Wallat, Avishek Anand, and Adam Jatowt. Tempretriever: Fusion-based temporal dense passage retrieval for time-sensitive questions. In *Proceedings of the 19th ACM International Conference on Web Search and Data Mining*, pp. 5–15. Association for Computing Machinery, 2026. doi: 10.1145/3773966.3777938.
- Akari Asai, Zeqiu Wu, Yizhong Wang, et al. Self-rag: Learning to retrieve, generate, and critique through self-reflection. In *International Conference on Learning Representations*, 2024.
- Gen Dong, Yanjie Gao, Liquan Li, et al. Agent skills can be harmful: An empirical study of skill-induced failures in llm agents, 2026.
- Tianyi Guan, Yiding Wang, Haotong Yang, et al. Continualskillbench: Can llm agents truly evolve their capabilities?, 2026.
- Jiale Han, Austin Cheung, Yubai Wei, et al. Rag meets temporal graphs: Time-sensitive modeling and retrieval for evolving knowledge, 2025.
- Hanbing Liu, Lang Cao, and Yang Li. RAG or learning? understanding the limits of LLM adaptation under continuous knowledge drift in the real world. In *Findings of the Association for Computational Linguistics: ACL 2026*, pp. 11234–11252. Association for Computational Linguistics, 2026. doi: 10.18653/v1/2026.findings-acl.546. URL <https://aclanthology.org/2026.findings-acl.546/>.
- NOAA Climate Prediction Center. ENSO Diagnostic Discussion Archive. https://www.cpc.ncep.noaa.gov/products/expert_assessment/ENSO_DD_archive.php, 2026. First-party monthly discussions accessed 2026-08-22.
- Qianjun Pan, Yutao Yang, Junsong Li, Jie Zhou, Kai Chen, Xin Li, Qin Chen, and Liang He. Anything2skill: Compiling external knowledge into reusable skills for agents, 2026.
- U.S. Bureau of Economic Analysis. GDP News Release Archive. <https://www.bea.gov/news/archive>, 2026. First-party historical releases accessed 2026-08-22.
- U.S. Energy Information Administration. Weekly Petroleum Status Report Archive. <https://www.eia.gov/petroleum/supply/weekly/archive/>, 2026. First-party weekly release archive accessed 2026-08-22.
- Xiaoman Wang, Yaoze Zhang, Wenzhuo Fan, et al. Ia-rag: Interval-algebra-driven temporal reasoning for dynamic knowledge retrieval, 2026a.
- Yixuan Wang, Yiyang Zhou, Yiming Liang, et al. Not all skills help: Measuring and repairing agent knowledge, 2026b.

- Tianxin Wei, Zhan Shi, Minhua Lin, et al. Evo-harness: Context-to-harness skill compilation for self-evolving agents, 2026.
- Ruiyao Xu, Tiankai Yang, and Wei-Chieh Huang. Hyperskill: Self-evolving llm agents via hypergraph-structured skill memory, 2026.
- Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, and Zhen-Hua Ling. Corrective retrieval augmented generation, 2024.
- Qingren Yao, Yaxuan Kong, Yuqi Nie, Yichen Li, Anna Vettoruzzo, Qingsong Wen, Ming Jin, Joaquin Vanschoren, and Stefan Zohren. Timesage-ev: A live benchmark for agentic time series analysis in evolving environments, 2026.
- Siyue Zhang, Yuxiang Xue, Yiming Zhang, Xiaobao Wu, Anh Tuan Luu, and Chen Zhao. MRAG: A modular retrieval framework for time-sensitive question answering. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pp. 3080–3118. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.findings-emnlp.167. URL <https://aclanthology.org/2025.findings-emnlp.167/>.
- Mingxuan Zheng, Yujin Zhou, Chuxue Cao, et al. Skillprox: Self-evolving agent skills via proximal textual gradient descent, 2026.
- Xiaolin Zhou, Jinbo Liu, Li Li, et al. Counterfactual trace auditing of llm agent skills, 2026.

A REPRODUCIBILITY AND SENSITIVITY AUDITS

T1/G1 are byte-identical across the initial replay and EIA validation. The provenance-clean r3 reaudit is 6221016166b71ebe; r4 adjudication is 316ee89a8eb1d53f. The r5 replication bundle contains executable r3/r4 endpoints, harnesses, scorers, runner scripts, frozen helper source, first-party archived pages, retained result JSON, and a SHA-256 manifest. This bundle reproduces the source-native study and remains separate from the unavailable TimeSage-EV evaluated snapshot.

A.1 EXACT FROZEN REUSABLE HELPERS

The six helper implementations below are the executed source, not pseudocode. Each targeted/generic pair has equal lexical-token count and equal AST-node count under the pre-execution audit.

```
# T1_temporal_cutoff.py
def skill(package, context):
    limit = context.get("cutoff_date", "")
    chosen = []
    for item in package.get("evidence_metadata", []):
        marker = item.get("release_date", "")
        ref = item.get("evidence_ref", "")
        if isinstance(marker, str) and marker <= limit:
            chosen.append(ref)
    return {"evidence_refs": chosen}

# G1_temporal_cutoff.py
def skill(package, context):
    limit = context.get("schema_ceiling", "")
    chosen = []
    for item in package.get("evidence_metadata", []):
        marker = item.get("schema_version", "")
        ref = item.get("evidence_ref", "")
        if isinstance(marker, str) and marker <= limit:
            chosen.append(ref)
    return {"evidence_refs": chosen}

# T2_release_alignment.py
import re
def skill(package, context):
    text = " ".join(package.get("document_text", "").split())
    match = re.search(r"([1-4])?(?:st|nd|rd|th) Quarter(?: and Year)? (20\d{2})",
        text, re.I)
    value = None
    if match:
        value = f"{match.group(2)}-Q{match.group(1)}"
```

```

594         return {"value": value}
595
596 # G2_release_alignment.py
597 import re
598 def skill(package, context):
599     text = " ".join(package.get("document_text", "").split())
600     match = re.search(r"(News) (Release|Archive|Statement) (?: and Data)? "
601                      r"([A-Za-z]{2,20})", text, re.I)
602     value = None
603     if match:
604         value = f"{match.group(1)}-{match.group(3)}"
605     return {"value": value}
606
607 # T3_exogenous_grounding.py
608 def skill(package, context):
609     markers = ("contributors to", "reflected", "because", "collectively",
610               "favored", "favors ", "associated with", "due to")
611     chosen = []
612     for span in package.get("candidate_spans", []):
613         text = span.get("text", "").lower()
614         sid = span.get("span_id", "")
615         if any(marker in text for marker in markers):
616             chosen.append(sid)
617     return {"span_ids": chosen}
618
619 # G3_exogenous_grounding.py
620 def skill(package, context):
621     markers = ("table", "figure", "release", "appendix", "contact",
622               "notes ", "technical notes", "page")
623     chosen = []
624     for span in package.get("candidate_spans", []):
625         text = span.get("text", "").lower()
626         sid = span.get("span_id", "")
627         if any(marker in text for marker in markers):
628             chosen.append(sid)
629     return {"span_ids": chosen}

```

A.2 EXPLORATORY CAPACITY-STRESS MATRIX

Split	Family	N0	Generic	Target	Δ_{T-G}	Δ_{T-N}
C3-R	Cutoff	32	0	72	+72	+40
C3-R	Align	100	100	100	0	0
C3-R	Ground	40	48	64	+16	+24
C4-R	Cutoff	53	13	100	+87	+47
C4-R	Align	80	93	100	+7	+20
C4-R	Ground	60	52	64	+12	+4

Table 2: Exploratory mini-model capacity stress over five repeats (percent), reported separately from the registered primary and replication tracks.

A.3 ENDPOINT-LEVEL UNCERTAINTY AND TEST RESOLUTION

Table 3 averages repeats within endpoint-condition and bootstraps endpoints; W/T/L is targeted versus generic, and sign tests use non-ties. Repeats never increase the inferential sample size. For n non-tied endpoints, the smallest attainable two-sided sign-test value is 2^{1-n} when every endpoint moves in the same direction. Thus two non-tied endpoints cannot yield a value below .5; three cannot go below .25; five cannot go below .0625; eight can reach .0078. Ties reduce resolution further. This arithmetic is why the two-endpoint grounding-transfer result is explicitly descriptive, why the three-endpoint Kimi alignment result remains underpowered despite a large effect, and why the eight-endpoint EIA validation supports a stronger endpoint-level claim.

A.4 WHY THE FAMILY ESTIMAND IS PRIMARY

The family contract and the full-task conjunctive score answer different questions. The family score asks whether the targeted temporal operation was corrected. The conjunctive score additionally requires orthogonal output fields and exact surface forms. A zero conjunctive score can therefore coexist with a successful cutoff intervention when temporal admissibility is correct but an unrelated

Layer	Split	Fam.	Δ_{T-G} [95%]	W/T/L	p_G	Δ_{T-N} [95%]	p_N
D	C3	Cut	28 [8,56]	4/1/0	.125	0 [0,0]	1
D	C3	Grd	40 [0,80]	2/3/0	.50	40 [0,80]	.50
D	C4	Cut	13 [0,20]	2/1/0	.50	0 [0,0]	1
D	C4	Grd	20 [0,60]	1/4/0	1	20 [0,60]	1
K	C3	Grd	10 [0,20]	2/3/0	.50	10 [0,30]	1
K	C4	Align	66.7 [50,75]	3/0/0	.25	41.7 [25,75]	.25
K	C4	Grd	20 [0,60]	1/4/0	1	20 [0,60]	1
M	C3	Cut	72 [32,100]	4/1/0	.125	40 [8,72]	.25
M	C3	Grd	16 [-12,60]	1/3/1	1	24 [0,64]	.50
M	C4	Cut	86.7 [60,100]	3/0/0	.25	46.7 [20,100]	.25
M	C4	Align	6.7 [0,20]	1/2/0	1	20 [0,60]	1
M	C4	Grd	12 [0,28]	2/3/0	.50	4 [-32,36]	1

Table 3: Endpoint-level uncertainty for non-ceiling initial-replay cells. D=DeepSeek primary, K=Kimi replication, M=exploratory mini.

field differs, or with successful release alignment when a semantically identical period is formatted differently. The family definitions were frozen before execution, so selecting them is determined by the experimental question and its time order. The conjunctive score is preserved as a sensitivity view and is never used to erase an unfavorable family result.

Family	r3 full-task conjunctive score	Effect on claim
Grounding	Identical to family score	Unchanged
Cutoff	All cells collapse to 0	Family estimand remains primary
Alignment	All cells collapse to 0	Family estimand remains primary

Table 4: Sensitivity to the auxiliary full-task exact score. The two columns evaluate different estimands.

A.5 CALL-RATE, PROMPT-LENGTH, AND CONDITION-POSITION AUDIT

The retained source-native rows contain 442 no-skill, 442 generic, and 442 targeted observations. Helper execution is deterministic in the frozen harness: generic and targeted execute exactly once before the model answer in 442/442 rows each, whereas no-skill executes no helper in 0/442 rows. Thus targeted-minus-generic cannot be explained by a higher helper invocation rate; this experiment does not measure endogenous tool-use propensity. Across exact targeted/generic pairs, targeted prompts are shorter by 18.85 input tokens on average (302 shorter, 140 longer). Condition-position counts are 157/135/150 for no-skill, 145/155/142 for generic, and 140/152/150 for targeted across positions 0/1/2. Pooled 3×2 success-by-position tests give $p = .993$, $.657$, and $.914$ for no-skill, generic, and targeted respectively; targeted-minus-no-skill remains positive at all three positions. These are post-hoc zero-call diagnostics and do not substitute for a prospectively counterbalanced replication. The full audit is included in the supplement.

A.6 ISO DATE AND PERIOD-REPRESENTATION AUDIT

A zero-call static audit covers the 23 initial BEA/NOAA endpoints and the 12 post-review EIA endpoints. Across the serialized endpoint objects, all 55 occurrences of `cutoff_date` and all 176 occurrences of `release_date` are calendar-valid YYYY-MM-DD strings. Whenever both top-level and `skill_context` cutoff fields are present they agree exactly. T1 makes 88 observed release-versus-cutoff comparisons over the frozen endpoint packages; lexical string order and parsed-calendar order agree on all 88. The executed T1 source is byte-identical between the initial and EIA phases (SHA-256 `df1950ef3930...d56d522`). Reporting-period identities are a separate representation layer: the initial replay uses canonical month/quarter identities for relevant families, while EIA uses day-level data-ending dates. These reporting-period strings never enter T1’s release-date comparator and are adjudicated by the frozen family scorer.

A.7 IDENTIFICATION LADDER AND PROSPECTIVE CONTROLS

Table 5 separates the two executed estimands from two stronger comparisons that cannot be answered from the existing rows.

Question	Contrast	Status	Identified quantity
Repair original agent?	T-N	Executed	Incremental value of inserting the targeted procedure over the original agent.
Beat target-blind help?	T-G	Executed	Targeted-operation value relative to the frozen complexity-matched helper.
Beat neutral extra help?	T-G ₀	Not executed	Operation-specific value beyond a separately validated behavior-neutral helper.
Does container matter?	T-R	Not executed	Callable-skill value beyond an operation-matched retrieval-side implementation.

Table 5: Identification ladder. G₀ and R are prospective treatments, not inferred baselines. Current claims use only the executed T-N and T-G contrasts.

The latter two identification questions remain open. We specify their treatment surfaces here to prevent later outcome-driven redesign. These are prospective designs only: they contribute zero model outcomes and do not upgrade any claim in the current paper.

G₀: neutral-helper candidate. A future confirmatory harness would add a fourth arm that executes exactly one helper call through the same wrapper as T and G but performs an identity projection: it returns the admissible evidence object in its original order without reading cutoff dates, mapping reporting periods, selecting grounding events, filtering candidates, or adding task-specific facts. The source, input/output schema, serialization policy, endpoint set, condition order, and model configuration must be frozen before confirmatory outcomes. A static validity gate must verify that G₀ accesses none of the registered temporal fields and preserves the frozen evidence identity/order contract on every endpoint. Behavioral neutrality is then an empirical requirement, not an assumption: an equivalence margin, independent endpoint count, and stopping rule must be frozen before any new outcomes are observed. If G₀ materially helps or harms N under that prospective criterion, it cannot serve as the neutral control. If it passes neutrality, the strengthened operation claim requires T to improve over both N and G₀ on the same endpoint units; the existing G remains a target-blind stress control rather than the sole generic anchor.

R: operation-matched retrieval-side treatment. A future container comparison would apply the same frozen temporal operation upstream during evidence retrieval/context construction rather than exposing it as callable skill state. For each endpoint, R must use the same candidate evidence, operation source/hash or a mechanically verified equivalent, selected evidence references/span IDs, model configuration, and outcome scorer as T. The visible operation output must be content-equivalent, while the integration surface differs: R materializes the result in retrieved context without the callable-skill wrapper. Model-call budget, endpoint units, decoding, and condition order must be matched prospectively. The estimand is $\hat{\Delta}_{T-R}$. If T and R are indistinguishable at the frozen resolution, the supported conclusion remains operation value with no container advantage. If R outperforms T, callable-container superiority is refuted. Only a prospectively resolved T>R result could motivate a separate container-value claim. No such result exists in the present paper.

Every retained row binds endpoint, condition, repeat, skill hash, prompt hash, raw response hash, and provider response-ID hash. Kimi repeat 4 is excluded by hash provenance with no replacement. The exact TimeSage-EV replication remains separate from all source-native evidence.